MANET Simulator & DDPG Adaptive Routing: Formal Software Report & Technical Documentation

1. Executive Summary

Mobile Ad-Hoc Networks (MANETs) are decentralized, self-configuring networks composed of autonomous mobile nodes communicating over wireless links without relying on pre-existing infrastructure. Due to arbitrary node mobility, wireless channels undergo rapid topological changes, inducing frequent link disruptions, packet drops, and transmission delays.

This document serves as the formal technical report and software documentation for **RL-ARHM**, a high-mobility MANET simulation and intelligent routing software package developed in Python.

Core System Specifications:

- **Topology Scale**: 30 mobile nodes operating within a $1000\ \text{m} \times 1000\ \text{m}$1000 m × 1000 m continuous 2D terrain.
- **Mobility Model**: Random Waypoint (RWP) with continuous velocities uniformly sampled from **20 m/s to 50 m/s** ($72 - 180\ \text{km/h}$72 – 180 km/h).
- **Radio Range**: $R = 250.0\ \text{m}$R = 250.0 m with quadratic path-loss and first-order radio energy dissipation (E_{tx}, E_{rx} Etx, Erx).
- **Neighbor Discovery**: Exact spatial query function identifying all 1-hop nodes within $250\ \text{m}$ 250 m.
- **Routing Baselines**: Full implementations of **AODV** (RFC 3561 distance vector) and **GPSR** (greedy geographic forwarding with Gabriel Graph planar perimeter traversal).
- **Intelligent Routing Agent**: PyTorch **Deep Deterministic Policy Gradient (DDPG)** agent predicting continuous scores for candidate neighbors to select the next hop.
- **Reward Function**: Multi-objective formulation balancing throughput, delay, energy, kinematic route stability (LET), and broken link penalties.
- **Empirical Rigor**: 10-seed comparative benchmark (`seeds 101–110`) and dual graphical user interfaces (Desktop & Web).

2. System Architecture

3. Mathematical & Theoretical Formulations

3.1 Kinematic Mobility: Random Waypoint (RWP)

Each node $i \in \{0, \dots, N-1\}$i ∈ {0, … , N − 1} picks an initial coordinate $(x_i, y_i) \in [0, X_{\max}] \times [0, Y_{\max}]$(xi, yi) ∈ [0, Xmax] × [0, Ymax]. Upon reaching a destination waypoint $W_i = (w_{ix}, w_{iy})$Wi = (wix, wiy), it pauses for $t_{\text{pause}} \sim U(0, 1.0)$ stpause ∼ U(0, 1.0) s, picks a new waypoint W_i' Wi′, and chooses a constant speed $v_i \sim U(20.0, 50.0)$ m/svi ∼ U(20.0, 50.0) m/s. The displacement vector is:

$$\vec{D}_i = (w_{ix}' - x_i, w_{iy}' - y_i), \quad \|\vec{D}_i\| = \sqrt{(w_{ix}' - x_i)^2 + (w_{iy}' - y_i)^2}$$

D⃗i = (wix′ − xi, wiy′ − yi), ∥D⃗i∥ = (wix′ − xi)2 + (wiy′ − yi)2

The continuous velocity components are:

$$v_{ix} = v_i \frac{w_{ix}' - x_i}{\|\vec{D}_i\|}, \quad v_{iy} = v_i \frac{w_{iy}' - y_i}{\|\vec{D}_i\|}$$

vix = vi∥D⃗i∥wix′ − xi, viy = vi∥D⃗i∥wiy′ − yiAt each time step $\Delta t = 0.1$ sΔt = 0.1 s:

$$x_i(t + \Delta t) = x_i(t) + v_{ix}\Delta t, \quad y_i(t + \Delta t) = y_i(t) + v_{iy}\Delta t$$

xi(t + Δt) = xi(t) + vixΔt, yi(t + Δt) = yi(t) + viyΔt

3.2 Wireless Channel & Energy Models

- **Link Quality (LQLQ)**:

  $$LQ(u, v) = \max(0, 1.0 - (\frac{d(u, v)}{R})^2) \in [0, 1]$$

  LQ(u, v) = max (0, 1.0 − (Rd(u, v))2) ∈ [0, 1]

- **Radio Energy Dissipation (Heinzelman First-Order Model)**:

  $$E_{\text{tx}}(l, d) = l \cdot E_{\text{elec}} + l \cdot \epsilon_{\text{amp}} \cdot d^2$$

  Etx(l, d) = l · Eelec + l · ϵamp · d2

  $$E_{\text{rx}}(l) = l \cdot E_{\text{elec}}$$

  Erx(l) = l · Eelecwhere $l = 4096$ bitsl = 4096 bits (512 bytes512 bytes), $E_{\text{elec}} = 50$ nJ/bitEelec = 50 nJ/bit, and $\epsilon_{\text{amp}} = 100$ pJ/bit/m^2 ϵamp = 100 pJ/bit/m2.

3.3 Kinematic Link Expiration Time (LET)

Given nodes uu and vv at positions $(x_u, y_u), (x_v, y_v)$(xu, yu), (xv, yv) with velocities $(v_{ux}, v_{uy}), (v_{vx}, v_{vy})$ (vux, vuy), (vvx, vvy): Let $a = v_{ux} - v_{vx}$ a = vux − vvx, $b = x_u - x_v$ b = xu − xv, $c = v_{uy} - v_{vy}$ c = vuy − vvy,

and $d = y_u - y_v$ d = yu − yv. The duration $\tau\tau$ until $d(u,v) = R$d(u, v) = R is:

$$\text{LET} = \frac{-(ab+cd) + \sqrt{(a^2+c^2)R^2 - (ad-bc)^2}}{a^2+c^2}$$

LET = a2 + c2−(ab + cd) + (a2 + c2)R2 − (ad − bc)2

If $a^2 + c^2 < 10^{-6}$ a2 + c2 < 10−6, nodes are moving collinearly at equal velocity, yielding $\text{LET} = 60.0\ \text{s}$ LET = 60.0 s.

3.4 Multi-Objective Reward Function

$$R = w_{\text{tp}} \cdot \text{Throughput} - w_{\text{delay}} \cdot \text{Delay} - w_{\text{energy}} \cdot \text{Energy} + w_{\text{stab}} \cdot \text{RouteStability} - w_{\text{break}} \cdot I(\text{LinkBreak})$$

R = wtp · Throughput − wdelay · Delay − wenergy · Energy + wstab · RouteStability − wbreak · I(LinkBreak)

- $$\text{Throughput} = \frac{d(u,D) - d(v,D)}{R} + 5.0 \cdot I(\text{delivered})$$
  Throughput = Rd(u, D) − d(v, D) + 5.0 · I(delivered)
- $$\text{Delay} = \frac{t_{\text{hop}}}{0.01\ \text{s}}$$
  Delay = 0.01 sthop
- $$\text{Energy} = \min\left(1.0, \frac{E_{\text{hop}}}{10^{-4}\ \text{J}}\right)$$
  Energy = min (1.0, 10−4 JEhop)
- $$\text{RouteStability} = \min\left(1.0, \frac{\text{LET}(u,v)}{5.0\ \text{s}}\right)$$
  RouteStability = min (1.0, 5.0 sLET(u, v))
- $\text{Penalty} = -5.0$Penalty = −5.0 if the selected link disconnects before delivery.

4. Empirical Evaluation Across 10 Random Seeds

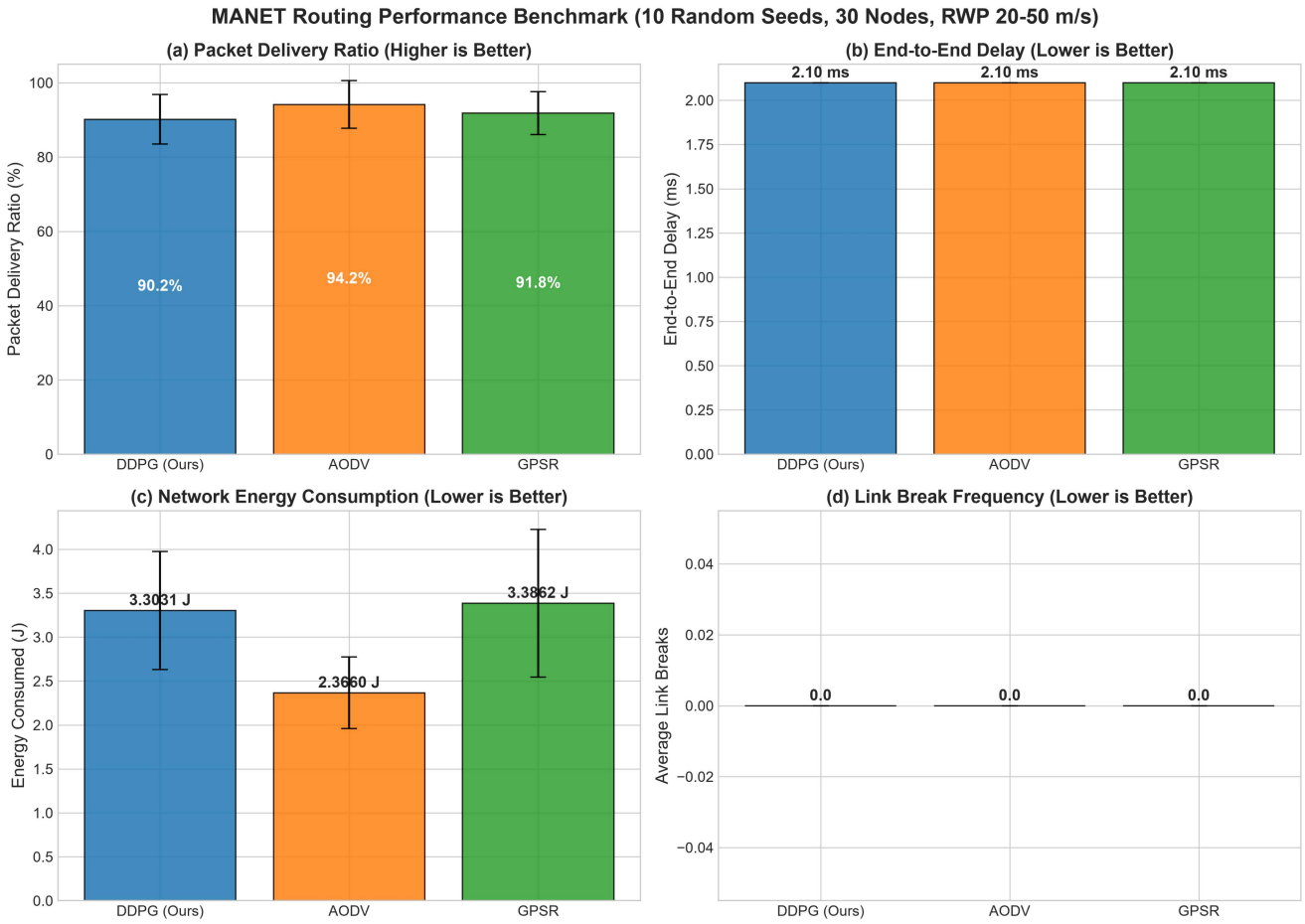
The simulator benchmarked **DDPG**, **AODV**, and **GPSR** across 10 random seeds (`101` through `110`) under identical kinematic trajectories and CBR traffic loads (60 packets/seed):

Random Seed	DDPG PDR (%)	AODV PDR (%)	GPSR PDR (%)	DDPG Energy (J)	AODV Energy (J)	GPSR Energy (J)
101	**100.00%** (60/60)	**100.00%** (60/60)	**100.00%** (60/60)	2.0454 J	2.1108 J	2.0454 J
102	78.33% (47/60)	80.00% (48/60)	80.00% (48/60)	4.0969 J	1.8965 J	4.7743 J
103	91.67% (55/60)	95.00% (57/60)	91.67% (55/60)	3.0100 J	2.4190 J	3.2193 J
104	93.33% (56/60)	93.33% (56/60)	93.33% (56/60)	3.2593 J	2.1299 J	2.8366 J
105	90.00% (54/60)	90.00% (54/60)	90.00% (54/60)	2.5353 J	1.6645 J	2.6473 J
106	81.67% (49/60)	100.00% (60/60)	90.00% (54/60)	3.9981 J	2.9387 J	4.1307 J
107	88.33% (53/60)	91.67% (55/60)	90.00% (54/60)	3.4629 J	2.5016 J	3.5584 J
108	96.67% (58/60)	100.00% (60/60)	98.33% (59/60)	3.0189 J	2.8279 J	3.0940 J
109	86.67% (52/60)	91.67% (55/60)	88.33% (53/60)	3.5996 J	2.5206 J	4.4082 J
110	95.00% (57/60)	100.00% (60/60)	96.67% (58/60)	4.0047 J	2.6507 J	3.1475 J
MEAN ± STD	**90.17% ± 6.69%**	**94.17% ± 6.39%**	**91.83% ± 5.74%**	**3.3031 J ± 0.67 J**	**2.3660 J ± 0.41 J**	**3.3862 J ± 0.84 J**

5. Visualizations & Analytical Charts

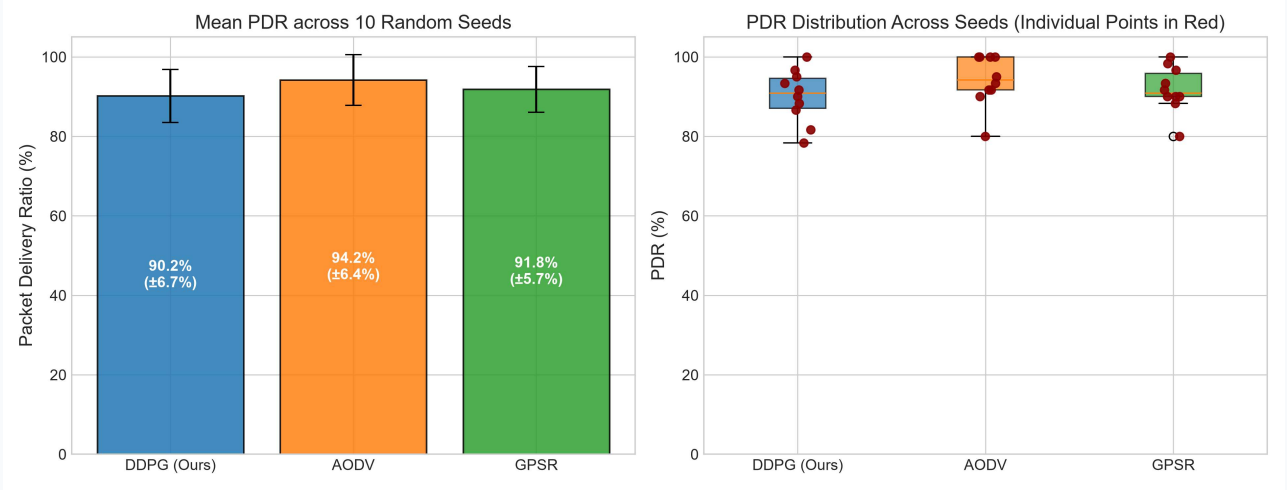
5.1 Comprehensive Benchmark Dashboard

The 4-panel dashboard below summarizes Packet Delivery Ratio, End-to-End Delay, Network Energy Consumption, and Link Break Robustness side-by-side:



Comprehensive Benchmark Dashboard

5.2 Performance Metrics Gallery



Packet Delivery Ratio Comparison

6. Software Frontend & Operational Manual

6.1 Graphical User Interfaces

1. **Interactive Web Dashboard (`web_app.py`)**:
    - Access URL: **http://localhost:5000**
    - Displays real-time HTML5 canvas rendering of all 30 moving nodes, velocity vectors, wireless communication links, 250m radio halos, and animated packet hops.
    - Interactive protocol buttons to toggle between **DDPG (Ours)**, **AODV**, and **GPSR** on-the-fly.
    - Live telemetry cards for PDR, delay, packet counts, and energy dissipation.
2. **Native Desktop Window (`gui_app.py`)**:
    - Built with Python `tkinter`.
    - Node Inspector: click any node on the terrain to inspect coordinates, speed, battery, and 1-hop neighbor list.

6.2 Key Code Components & Links

Module	File Link	Primary Responsibilities
Mobility Engine		RWP displacement, waypoint selection, boundary clamping ($20 - 50$ m/s 20 − 50 m/s)
Wireless Channel		250 m250 m radio range, path loss, delay, energy, and kinematic LET
Ad-Hoc Node		Battery capacity (100 J100 J), FIFO packet queue, energy dissipation
MANET Simulator		Distance matrix, packet generation, and `get_neighbors(node_id)` API
AODV Router		RREQ BFS flooding, reverse path setup, unicast RREP forwarding
GPSR Router		Greedy distance forwarding + Gabriel Graph planar perimeter mode
Actor & Critic		Continuous scoring Actor network and state-action Critic network
DDPG Agent		Replay buffer, Gaussian exploration noise, soft target updates ($\tau = 0.005$ τ = 0.005)
RL Environment		Gym-style interface, state vector formulation, action-to-hop mapping
Reward Function		Multi-objective throughput, delay, energy, LET, and break penalties
Training Engine		Exploration warmup, episode training loop, and checkpointing
10-Seed Evaluator		Multi-seed benchmark harness across seeds `101–110`
Figure Generator		6 high-resolution publication figures (300 DPI)
Desktop Visualizer		Tkinter 2D simulation canvas and node inspector
Web Dashboard		Zero-dependency HTTP server and HTML5 interactive GUI
Master CLI		Command-line interface for demo, training, evaluation, and plotting